ATTACHMENT

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5/27/10

Budget Revision Request 2 – El Estero WWTF FOG Pilot Project

On February 17, 2010, City Council authorized AECOM to proceed with the design of a pilot Fats Oil and Grease (FOG) receiving, handling, and injection station at the El Estero Wastewater Treatment Plant (WWTP). Our approved scope includes plans and technical specifications for the site piping, pumping, storage vessel, and local controls of the receiving and delivery system. We are also contracted to provide limited construction phase support.

City staff has requested AECOM to include additional project features and engineering tasks to our approved scope of work. The additional scope items requested are:

1. Design of a containment area surrounding the FOG processing area
2. Site drain to Cul-de-sac drain

Through our discussions with City staff, we have developed the following understanding of the requested modifications to our Scope of Work.

Additional Task Item 1 – FOG Containment Facilities

Done

AECOM will provide construction plans and specifications for a containment wall surrounding the FOG processing area. Preliminary estimates indicate a concrete wall (25' x 25' x 3' high) will be required to contain 150% of the volume of the FOG storage tank. The design will include stairs with a handrail to access the containment area, and the containment area will feature concrete flatwork within the entire contained area. Flatwork will be sloped to a sump. The sump will be connected to a manually operated valve to allow periodic draining of collected water to the cul-de-sac drain (see item 2 below). As discussed, pumps will be located within the contained area. Truck offload equipment (rock trap, hose connection, etc.) will be located outside the containment area. Per Staff request, our design will not feature a roof structure.

We have reviewed record drawing information provided by City staff and have not identified any conflict with existing utilities in the area of the proposed containment area. We assume that no such conflicts exist. The construction contractor will be responsible to identify and mitigate any conflicts identified during construction.

Additional Task Item 2 - Drainage Facilities

AECOM will provide construction plans and specifications for a gravity drain connection (assumed 6 inch) between the proposed containment area and the existing cul-de-sac drain. We will also connect the proposed FOG rock trap to the new drain to facilitate maintenance of the screen. Seal water from the positive displacement pumps will also be directed to the drain.

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We have assumed that adequate fall exists between the proposed containment area and the existing drain, and that no conflicts exist along the alignment. Through discussions with plant staff and our preliminary review of existing record drawings, we assume that the nearby utilities (such as the plant outfall line) are adequately deep as to not impact the design of the proposed drain line.

We have budgeted 6 hours of time to further review existing as-built drawings to evaluate conflicts and determine grade lines such as inverts at the existing cul-de-sac drain. If adequate cover does not exist over the proposed drain line, a slurry trench may be recommended.

Additional Task Item 3 – Requested Revisions to Existing Work

To date AECOM has performed work beyond what was approved in our original scope and budget (e.g. providing a reinforced concrete slab and seismic design, and participating in additional meetings and coordination efforts). We have also evaluated various revisions to the scope (additional digester piping, SCADA integration, and drainage improvements), analyzed potential impacts to project cost, and provided an opinion of cost for each potential revision.

Additionally, upon review of the Preliminary Design Memo (submitted April 28, 2010), significant modifications to the original scope of the project were requested by City staff. To incorporate these requests, AECOM must abandon some work already completed, revise other existing work, and perform new work not anticipated. These tasks include:

- Revising system layout
- Recalculating pump system curves
- Performing additional electrical calculations and layout
- Performing additional CAD design work
- Design of additional equipment pedestals
- Providing technical specs for a different tank and appurtenances,
- Revising cost opinions
- Providing additional direction for controlling and programming two duplex pump systems.

To accommodate the direction requested by City staff without eroding the approved budget for the remaining work, AECOM is requesting additional budget of $4,277 to incorporate these revisions. As always, work would be performed on a time-and-materials basis and will be billed based on actual work performed.

Additional Task Item 4 – Additional Construction Phase Support

Our approved Scope of Work includes limited construction phase support. We are currently authorized to respond to four (4) RFIs, assist the City with examining bids, attend a pre-construction meeting, and manage submittals. With the additional project components noted above, we are requesting an increase in budget for additional submittal review and RFI response.

We are also authorized to be onsite for four (4) hours during the first day of field work, four (4) hours during one progress visit during construction, and for four (4) hours for one final visit to evaluate the near completed facilities. Our site visits are intended to monitor the overall

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construction progress of the project and to address any major deficiencies encountered. Our approved scope does not include construction management or construction observation where daily activities are monitored and recorded. We are therefore not requesting additional budget to manage the additional construction activities. If unforeseen site conditions or quality concerns arise as construction proceeds, AECOM can perform additional support and site visits on an as-needed time and materials basis to assist with resolving issues and developing solutions in the field. Any additional services would be requested in writing for approval prior to the work being performed.

An hourly breakdown of these scope revisions can be seen on the attached worksheet (attached).

Budget Summary

Existing Approved Budget	Budget Revision Request 2	Revised Budget
$64,506	$21,859	$86,365

Please sign and return the enclosed copy of this letter, thereby enabling us to perform these additional services on a time and materials basis.

Thank you,

Jon Hanlon, PE

City of Santa Barbara

Signed:
Date:
Title:

Project Budget

El Estero FOG Pilot Program
Budget Revision 2

City of Santa Barbara

Task Description	Personnel Hours					Budget		
	Principal	Managing Engineer	Associate Engineer	Drafter	Total Hours	Labor	Non-Labor Fees	Total
Additional Task Item 1 - FOG spill Containment								
Plans (1 sheet)		1	4	12	17	$ 1,935	$ 155	$ 2,090
specs		1	8		9	$ 1,175	$ 94	$ 1,269
structural engineering	6				6	$ 1,200	$ 96	$ 1,296
Misc. details		2	8	8	18	$ 2,190	$ 175	$ 2,365
Subtotal	**6**	**4**	**20**	**20**	**50**	**$ 6,500**	**$ 520**	**$ 7,020**
Additional Task Item 2 - Drainage Facilities								
record drawing review			6		6	$ 750	$ 60	$ 810
Plans (1 sheet)		2	6	12	20	$ 2,360	$ 189	$ 2,549
Specs		2	8		10	$ 1,350	$ 108	$ 1,458
Subtotal	**-**	**4**	**20**	**12**	**36**	**$ 4,460**	**$ 357**	**$ 4,817**
Task Group 3 -								
Revisions to Work Already Completed		4	16	12	32	$ 3,960	$ 317	$ 4,277
Subtotal	**-**	**4**	**16**	**12**	**32**	**$ 3,960**	**$ 317**	**$ 4,277**
Task Group 4 - Additional Construction Phase Services								
RFIs and Bid Addenda		2	4		6	$ 850	$ 68	$ 918
Submittal Review and Management		4	16		20	$ 2,700	$ 216	$ 2,916
Change Order Review and Management		2	8		10	$ 1,350	$ 108	$ 1,458
additional record drawings				4	4	$ 420	$ 34	$ 454
Subtotal	**-**	**8**	**28**	**4**	**40**	**$ 5,320**	**$ 426**	**$ 5,746**
Total	**6**	**20**	**84**	**48**	**158**	**$ 20,240**	**$ 1,619**	**$ 21,859**

Amounts shown are fee.

Personnel Category	$/HR
Principal	$200.00
Managing Engineer	$175.00
Associate Engineer	$125.00
Drafter	$105.00